# HW #9

ALGEBRAIC NUMBER THEORY, GU4043; INSTRUCTOR: GYUJIN OH

**Question 1.**

(1) Find all integers $a \in \mathbb{Z}$ such that $5X^2 = a$ has a solution in $\mathbb{Z}_5 = \mathcal{O}_{\mathbb{Q}_5}$.

(2) Show that $f(X) = X^4 - 2X^3 - X^2 + 6$ is irreducible in $\mathbb{Q}[X]$.

**Hint.** Use the 2-adic Newton polygon (Theorem 4.43) and Gauss's lemma to deduce that, if $f(X)$ is not irreducible, $f(X) = g(X)h(X)$ for $g(X), h(X) \in \mathbb{Z}[X]$ with $\deg g = \deg h = 2$. Deduce a contradiction by combining information you can obtain by applying Hensel's lemma mod 2 and mod 3.

*Proof.*

(1) By the valuation reason, $a$ must be exactly divisible by an odd power of 5, so $a = 5^{2n+1}b$ for $n \geq 0$ and $(b, 5) = 1$. Then, the question is about $X^2 = b$ in $\mathbb{Z}_5$. By Hensel's lemma, this has a solution if and only if it has a solution mod 5, so whenever $b \equiv 1, 4 \pmod 5$.

(2) Suppose $f$ is reducible. Suppose that $f$ factorizes into degree 3 times degree 1. By Gauss lemma, $f$ has an integer root, necessarily dividing 6. The 2-adic Newton polygon implies that in $\mathbb{Q}_2[X]$, it factorizes into either $f(X) = g(X)h(X)$ with $\deg g = \deg h = 2$ or $f(X) = g(X)h(X)i(X)$ with $\deg g = 2$ and $\deg h = \deg i = 1$. The factorization of $f$ in $\mathbb{Q}[X]$ implies that the integer root of $f$ must have 2-adic valuation 0, or that it is odd. Thus, it must be either $\pm 1$ or $\pm 3$. Checking them,

$$f(\pm 1) = 1 \mp 2 - 1 + 6 = 6 \mp 2 \neq 0, f(\pm 3) = 81 \mp 54 - 9 + 6 = 78 \mp 54 \neq 0,$$

so a contradiction. Thus $f(X) = g(X)h(X)$ with $g, h \in \mathbb{Z}[X]$ and $\deg g = \deg h = 2$. The 2-adic Newton polygon analysis implies that one of $g, h$ is Eisenstein, and the other has Newton polygon the straight line with slope 0, so without loss of generality, let's say $g(X)$ is Eisenstein. This means that if $g(X) = X^2 + aX + b$, then $2|a$, $2|b$, 4 doesn't divide $b$. Similarly, if $h(X) = X^2 + cX + d$, then $d$ is odd. So we just say $g(X) = X^2 + 2aX + 2b$. On the other hand, $f(X)$ mod 3 can factorize into degree 2 times degree 2 in only one way, $X^2(X^2 - 2X - 1)$, so either $h(X) \equiv X^2 \pmod 3$ and $g(X) \equiv X^2 - 2X - 1 \pmod 3$ or the other way around.

- If $h(X) \equiv X^2 \pmod 3$ and $g(X) \equiv X^2 - 2X - 1 \pmod 3$, then $c, d$ are divisible by 3, and $2b \equiv -1 \pmod 3$, so $2b = 2$ and $d = 3$. Then $h(X) = X^2 + 3eX + 3$ and $g(X) = X^2 + 2aX + 2$ where $a \equiv -1 \pmod 3$. But then the $X$-term of $g(X)h(X)$ is $0 = 6eX + 6aX$, so $a = -e$. Then the $X^3$-term of $g(X)h(X)$ is $2aX^3 + 3eX^3 = -aX^3 = -2X^3$, so $a = 2$. Then

  $$g(X)h(X) = (X^2 + 4X + 2)(X^2 - 6X + 3) = X^4 - 2X^3 - 19X^2 + 6,$$

  contradiction.
- If $h(X) \equiv X^2 - 2X - 1 \pmod 3$ and $g(X) \equiv X^2 \pmod 3$, then $g(X) = X^2 + 6eX + 6f$, and $d \equiv -1 \pmod 3$, so $f = d = -1$. Thus we have $g(X) = X^2 + 6eX - 6$

and $h(X) = X^2 + cX - 1$, with $c \equiv -2 \pmod 3$. The $X$-term of $g(X)h(X)$ is $-6eX - 6cX = 0$, so $c = -e$. The $X^3$-term is $6eX^3 + cX^3 = -2X^3$, so it is $5eX^3$, which is impossible. Thus a contradiction.

All in all, a contradiction. Thus $f$ is irreducible in $\mathbb{Q}[X]$.

$\square$

**Question 2.** This generalizes HW#7, Question 2.

Let $n > 1$ be an integer, and let $K$ be a number field that contains $\mathbb{Q}(\zeta_n)$. Let $\alpha \in K^\times$. Suppose that $\alpha$ is not an $\ell$-th power in $K$ for any prime divisor $\ell$ of $n$ (i.e., for each prime divisor $\ell$ of $n$, there is no $x \in K$ such that $\alpha = x^\ell$). Consider $L = K(\sqrt[n]{\alpha})$. Let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_K$, and let $p$ be the rational prime that $\mathfrak{p}$ divides.

(1) Show that $X^n - \alpha$ is irreducible in $K[X]$, so that $[L : K] = n$.

**Hint.** For a monic polynomial $h(X)$ dividing $X^n - \alpha$, its roots are all of the form $\zeta_n^j \sqrt[n]{\alpha}$. Look at the constant term of $h(X)$.

(2) Show that $L/K$ is Galois, with $\mathrm{Gal}(L/K) \cong \mathbb{Z}/n\mathbb{Z}$.

(3) If $v_\mathfrak{p}(\alpha)$ is not divisible by $n$, show that $\mathfrak{p}$ is ramified in $L$.

(4) Suppose that $(n, p) = 1$. If $v_\mathfrak{p}(\alpha)$ is divisible by $n$, show that $\mathfrak{p}$ is unramified in $L$.

**Hint.** Use the relative discriminant.

(5) More precisely, in (3), show that $f, g$ can be described as follows. Take $\alpha = \pi^{nm}u$, where $\pi$ is a uniformizer of $K_\mathfrak{p}$, $m \in \mathbb{Z}$, and $v_\mathfrak{p}(u) = 0$. Consider the prime factorization of $X^n - u$ in $k_{K_\mathfrak{p}}[X]$, where $k_{K_\mathfrak{p}}$ is the residue field of $K_\mathfrak{p}$, written as

$$X^n - u = \overline{h}_1(X) \cdots \overline{h}_m(X), \quad \overline{h}_1(X), \cdots, \overline{h}_m(X) \in k_{K_\mathfrak{p}}[X].$$

Then, $g = m$, and $f = \frac{n}{m} = \deg \overline{h}_1 = \cdots = \deg \overline{h}_m$.

**Hint.** Show that $\zeta_n^j \not\equiv 1 \pmod{\mathfrak{p}}$ for $j = 1, \cdots, n-1$. Use this and Hensel's lemma to show that the prime fatcorization of $X^n - u$ in $K_\mathfrak{p}[X]$ is given by $X^n - u = h_1(X) \cdots h_m(X)$, where $\deg h_i = \deg \overline{h}_i$, and $h_i$ is a lift of $\overline{h}_i$. Use (1) and Theorem 4.53 to finish.

(6) Suppose that $p|n$, and let $k = v_p(n)$ (i.e., $p^k$ divides $n$ and $p^{k+1}$ does not). Suppose that $v_\mathfrak{p}(\alpha)$ is divisible by $n$, so that $\alpha = \pi^{nm}u$ for a uniformizer $\pi$ of $K_\mathfrak{p}$, $m \in \mathbb{Z}$, and $v_\mathfrak{p}(u) = 0$. Finally, let $s$ be the smallest integer strictly larger than $\frac{e_{K_\mathfrak{p}/\mathbb{Q}_p}}{p-1}$. Show that $\mathfrak{p}$ is unramified in $L$ if $u \in \mathcal{O}_{K_\mathfrak{p}}$ is a $p^k$-th power modulo $\pi^s\mathcal{O}_{K_\mathfrak{p}}$.

**Hint.** Let $n = p^k n'$, $(p, n') = 1$. Use Exercise 4.4 to show that any element of $1 + \pi^s\mathcal{O}_{K_\mathfrak{p}}$ is a $p^k$-th power. Use Theorem 4.53 to show that $\mathfrak{p}$ is unramified in $K(\sqrt[p^k]{\alpha})$ if $\alpha$ is a $p^k$-th power in $K_\mathfrak{p}$. Use (5) to show that $\mathfrak{p}$ is unramified in $K(\sqrt[n']{\alpha})$.

*Proof.*

(1) Suppose that a monic polynomial $h(X) \in K[X]$ divides $X^n - \alpha$, with $\deg h > 0$. Then, its roots, which could be all found in $L$, are $\{\zeta_n^{j_1} \sqrt[n]{\alpha}, \cdots, \zeta_n^{j_m} \sqrt[n]{\alpha}\}$ for some $0 \le j_1 < \cdots < j_m < n$. Therefore, the constant term of $h(X)$ is $\zeta_n^{j_1 + \cdots + j_m} \alpha^{\frac{m}{n}}$. For this to be in $K$, it must be that $m = n$, hence making $X^n - \alpha$ irreducible.

(2) Obvious.
(3) Same proof as HW#7, Question 2.
(4) Same proof as HW#7, Question 2.
(5) We can assume that $v_{\mathfrak{p}}(\alpha) = 0$. From Theorem 4.53, we know that $L \otimes_K K_{\mathfrak{p}} \cong \prod_{\mathfrak{q}|\mathfrak{p}} L_{\mathfrak{q}}$. Note that $e$'s and $f$'s are the same throughout the primes dividing $\mathfrak{p}$ as $L/K$ is Galois, and $e = 1$ by (4). Therefore, it suffices to know what $[L_{\mathfrak{q}} : K_{\mathfrak{p}}]$ is for a single $\mathfrak{q}|\mathfrak{p}$, and then $f = [L_{\mathfrak{q}} : K_{\mathfrak{p}}]$ and $g = \frac{n}{f}$. If $X^n - \alpha = g_1(X) \cdots g_t(X)$ is the prime factorization of $X^n - \alpha$ in $K_{\mathfrak{p}}[X]$, then, by Chinese Remainder Theorem,

$$L \otimes_K K_{\mathfrak{p}} = L[X]/(X^n - \alpha) \otimes_K K_{\mathfrak{p}} = K_{\mathfrak{p}}[X]/(X^n - \alpha) \cong \prod_{i=1}^{t} K_{\mathfrak{p}}[X]/g_i(X).$$

Therefore, this means that $f = \deg g_1 = \cdots = \deg g_t$.

By Gauss's lemma, $g_i(X) \in \mathcal{O}_K[X]$. Let $\overline{g}_i(X) \in k_{K_{\mathfrak{p}}}[X]$ be the reduction mod $\mathfrak{p}$ of $g_i(X)$. Then, $X^n - \alpha = \overline{g}_1(X) \cdots \overline{g}_t(X)$ is a factorization in $k_{K_{\mathfrak{p}}}[X]$. We want to show that this is the prime factorization. It then suffices to show that the length of the prime factorization of $X^n - \alpha$ is exactly $t$ (here, length means the number of prime factors appearing in the prime factorization, counted with multiplicities). We know that the length is at least $t$ because of the expression $X^n - \alpha = \overline{g}_1(X) \cdots \overline{g}_t(X)$. Therefore, it suffices to show that a prime factorization $X^n - \alpha = \overline{h}_1(X) \cdots \overline{h}_m(X)$ lifts to a prime factorization $X^n - \alpha = h_1(X) \cdots h_m(X)$ in $\mathcal{O}_{K_{\mathfrak{p}}}[X]$. For this, one only needs to check that one can use Hensel's lemma repeatedly, which is to say that we need to check that $\overline{h}_i(X)$'s are coprime to each other. This is the same as $X^n - \alpha$ being separable as a polynomial in $k_{K_{\mathfrak{p}}}[X]$, which is true as $(X^n - \alpha, nX^{n-1}) = 1$.
(6) Again, we can assume that $v_{\mathfrak{p}}(\alpha) = 0$. Note that $\mathfrak{p}$ is unramified in $K(\sqrt[n]{\alpha})$ by (5). Therefore, to show that $\mathfrak{p}$ is unramified in $L = K(\sqrt[n]{\alpha})$, it suffices to show that it is unramified in $K(\sqrt[p^k]{\alpha})$, as $L$ is the compositum of $K(\sqrt[n]{\alpha})$ and $K(\sqrt[p^k]{\alpha})$. Thus we can assume that $n = p^k$. Theorem 4.53 says that $\mathfrak{p}$ in fact splits completely in $L$ if $\alpha$ is an $n$-th power in $K_{\mathfrak{p}}$, which is what we will show.

We know that $\alpha$ is a $p^k$-th power modulo $\pi^{ke_{K_{\mathfrak{p}}/\mathbb{Q}_p}+\epsilon}\mathcal{O}_{K_{\mathfrak{p}}}$. Thus, by dividing by an appropriate $p^k$-th power, it suffices to assume that $\alpha \equiv 1 \pmod{\pi^{ke_{K_{\mathfrak{p}}/\mathbb{Q}_p}+\epsilon}}$. On the other hand, by Exercise 4.4, $1+\pi^{ke_{K_{\mathfrak{p}}/\mathbb{Q}_p}+\epsilon}\mathcal{O}_{K_{\mathfrak{p}}}$ as a multiplicative group is isomorphic to $\pi^{ke_{K_{\mathfrak{p}}/\mathbb{Q}_p}+\epsilon}\mathcal{O}_{K_{\mathfrak{p}}}$ as an additive group. Here, we are using that

$$ke_{K_{\mathfrak{p}}/\mathbb{Q}_p} + \epsilon > \frac{e_{K_{\mathfrak{p}}/\mathbb{Q}_p}}{p-1}$$

for any $p$ (which is why we need $\epsilon$ term for $p = 2$). As $p^k$ divides $\pi^{ke_{K_{\mathfrak{p}}/\mathbb{Q}_p}+\epsilon}$, any element in $1 + \pi^{ke_{K_{\mathfrak{p}}/\mathbb{Q}_p}+\epsilon}\mathcal{O}_{K_{\mathfrak{p}}}$ is a $p^k$-th power.

□

**Question 3.** Let $K$ be a number field, and let $\mathfrak{p}$ be a prime ideal of $\mathcal{O}_K$. For any positive integers $N, M > 0$, show that there exists a number field $L$ over $K$ which has a prime ideal $\mathfrak{q}$ that divides $\mathfrak{p}$ such that $e(\mathfrak{q}|\mathfrak{p}) \geq N$, $f(\mathfrak{q}|\mathfrak{p}) \geq M$.

**Hint.** Use Krasner's lemma, HW#8, Question 2(2).

*Proof.* There is definitely a local field $F$, a finite extension of $K_\mathfrak{p}$, such that $e_{F/K_\mathfrak{p}} = N$ and $f_{F/K_\mathfrak{p}} = M$. For example, first take $G/K_\mathfrak{p}$ to be the unramified extension of $K_\mathfrak{p}$ of degree $M$, and then take the totally ramified extension $F/G$ of degree $N$ that is obtained by adjoining a root of an Eisenstein polynomial of degree $N$. By multiplicativity of $e$ and $f$ for a tower of local fields, we can $e_{F/K_\mathfrak{p}} = N$ and $f_{F/K_\mathfrak{p}} = M$.

Thus it suffices to show the existence of $L$ and $\mathfrak{q}|\mathfrak{p}$ such that $L_\mathfrak{q} \supset F$. Let $F = K_\mathfrak{p}(\alpha)$ for some $\alpha \in F$, and let $f(X) \in K_\mathfrak{p}[X]$ be the minimal polynomial of $\alpha$ over $K_\mathfrak{p}$. Let $\alpha_1 = \alpha, \alpha_2, \cdots, \alpha_n$ be the roots of $f(X)$ in a large enough local field $E/F$. Let $\epsilon > 0$ be such that $\epsilon < |\alpha_i - \alpha_j|$ for any $1 \leq i < j \leq n$.

Let $\epsilon' > 0$ be a small number to be determined. If $f(X) = (X - \alpha_1) \cdots (X - \alpha_n) = X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0$, then there exist $b_{n-1}, \cdots, b_0 \in K$ such that $|a_i - b_i| < \epsilon'$; this is because $K_\mathfrak{p}$ is the completion of $K$, so $K$ is dense in $K_\mathfrak{p}$. Let $D/E$ be a large enough local field such that $g(X) = X^n + b_{n-1}X^{n-1} + \cdots + b_1 X + b_0$ has roots $\beta_1, \cdots, \beta_n \in D$. Then, one one hand, for any $1 \leq i \leq n$,

$$|g(\alpha_i)| = \left| \sum_{j=0}^{n-1} (a_j - b_j)\alpha_i^j \right| \leq \epsilon' \max\{1, |\alpha_i|^{n-1}\}.$$

On the other hand, $|g(\alpha_i)| = \prod_{j=1}^n |\alpha_i - \beta_j|$. Now, we let $\epsilon' < \frac{\epsilon^n}{\max\{1, |\alpha_1|^{n-1}, \cdots, |\alpha_n|^{n-1}\}}$. Then, for any $1 \leq i \leq n$, $|g(\alpha_i)| < \epsilon^n$. This implies that, for any $1 \leq i \leq n$, there exists $1 \leq j_i \leq n$ such that $|\alpha_i - \beta_{j_i}| < \epsilon$.

Suppose that, for $i_1 \neq i_2$, $j_{i_1} = j_{i_2}$. Then, we have

$$\epsilon < |\alpha_{i_1} - \alpha_{i_2}| \leq \max(|\alpha_{i_1} - \beta_{j_{i_1}}|, |\alpha_{i_2} - \beta_{j_{i_2}}|) < \epsilon,$$

which is a contradiction. Therefore, $i \mapsto j_i$ is a permutation of $\{1, \cdots, n\}$. This implies that, for $i \neq k$, $|\alpha_k - \beta_{j_i}| \geq \epsilon$. By Krasner's lemma, $K_\mathfrak{p}(\beta_{j_1}) \supset K_\mathfrak{p}(\alpha) = F$. As $\beta_{j_1} \in \overline{\mathbb{Q}}$, we can take $L = K(\beta_{j_1})$. $\square$